

Exam type problems to solve in the last lecture

1. How many solutions has the following problem: a) $x'' + t^2x = 0$, $x(0) = 0$? b) $x'' + t^2x = 0$, $x(0) = 0$, $x'(0) = 0$? c) $x'' + t^2x = 0$; $x(0) = 0$, $x'(0) = 0$, $x''(0) = 1$? $\diamond$

2. Let $v \in C^1(\mathbb{R})$ with $v(0) = 0$, $v'(0) \neq 0$, şi $xv(x) > 0$ pentru orice $x \in \mathbb{R}^*$. We consider the planar system

$$\dot{x} = y, \quad \dot{y} = -v(x).$$

(a) Find a first integral in $\mathbb{R}^2$.

(b) Study the stability of the equilibrium point $(0, 0)$. $\diamond$

3. We consider the prey-predator system

$$\dot{x} = x(1 - y), \quad \dot{y} = -y(2 - x).$$

(a) Find the expression of a first integral in $(0, \infty) \times (0, \infty)$. Check it using the corresponding first order partial differential equation.

(b) If $(2, 1)$ is an equilibrium point, is it hyperbolic? $\diamond$

4. We consider the following nonlinear planar systems

$$\dot{x} = -x + xy, \quad \dot{y} = -2y + 3y^2.$$

a) Find its equilibria and study their stability.

b) Find $\varphi(t, 0, 2/3)$, $\varphi(t, 4, 0)$ and $\varphi(t, 1, 2/3)$.

c) Represent in the phase plane the orbits corresponding to the initial values $(0, 2/3)$, $(4, 0)$ and $(1, 2/3)$. $\diamond$

5. Let $\omega, \mu > 0$ be fixed parameters. The second order nonlinear equation $\theta'' + \mu\theta' + \omega^2 \sin \theta = 0$ describes the oscillations of a simple pendulum. Justify that the equilibrium solution $\theta(t) = 0$ for all $t \in \mathbb{R}$ is an attractor. $\diamond$

6. Let $\omega > 0$ be a fixed parameter. The second order linear equation $\theta'' + \omega^2\theta = 0$ describes the *small* oscillations of a simple idealized pendulum. Here $\theta(t)$ is the angle (measured in radians) between the rod and the vertical and t is the time variable measured in minutes. Find the solution of the IVP

$$\theta'' + \omega^2\theta = 0, \quad \theta(0) = \frac{\pi}{6}, \quad \theta'(0) = 0.$$

Describe the oscillations of the pendulum. After how much time the pendulum will return to the initial state? Represent the phase portrait of the associated planar dynamical system. Specify its type. Justify that the equilibrium solution $\theta(t) = 0$ for all $t \in \mathbb{R}$ is stable, but it is not an attractor. $\diamond$

7. Let $\omega > 0$ be a fixed parameter. The second order nonlinear equation $\theta'' + \omega^2 \sin \theta = 0$ describes the oscillations of a simple idealized pendulum.

(a) Find the equilibria. Use the linearization method to study their stability.

(b) Find a global first integral. Justify that the equilibrium solution $\theta(t) = 0$ for all $t \in \mathbb{R}$ is stable, but it is not an attractor. $\diamond$

8. We consider the nonlinear planar system

$$\dot{x} = -y - x(x^2 + y^2), \quad \dot{y} = x - y(x^2 + y^2).$$

(a) Justify that the equilibrium point $(0, 0)$ is not hyperbolic.

(b) Represent the phase portrait (using polar coordinates).

(c) Reading the phase portrait, deduce that the equilibrium point $(0, 0)$ is an attractor. $\diamond$

9. We consider the nonlinear planar system

$$\dot{x} = -y + x(1 - x^2 - y^2), \quad \dot{y} = x + y(1 - x^2 - y^2).$$

(a) Check that $\varphi(t, 1, 0) = (\cos t, \sin t)$ for all $t \in \mathbb{R}$.

(b) Represent the phase portrait (using polar coordinates).

(c) Reading the phase portrait, deduce that the equilibrium point $(0, 0)$ is a repeller. There is an attractor in this phase portrait? $\diamond$

10. We consider the linear planar system

$$\dot{x} = -x + 2y, \quad \dot{y} = 2x - 4y.$$

Find its general solution using two methods:

- (a) the reduction method;
- (b) the characteristic equation method. $\diamond$

11. Let $D: C^1(\mathbb{R}) \rightarrow C(\mathbb{R})$ be defined by $D(x) = x'$. Let $l(r) = (r + 2)^3$ and $\mathcal{L}: C^5(\mathbb{R}) \rightarrow C(\mathbb{R})$ be $\mathcal{L} = l(D)$.

- (a) Compute $D(e^{-t})$, $(D + 2)(e^{-t})$ and $\mathcal{L}(e^{-t})$.
- (b) Compute $D(e^{-2t})$, $(D + 2)(e^{-2t})$, $(D + 2)(te^{-2t})$, $(D + 2)(t^2e^{-2t})$ and $\mathcal{L}(t^3e^{-2t})$.
- (c) Let $A > 0$. Find the general solution of $\mathcal{L}(x) = Ae^{-t}$.
- (d) Let $A > 0$. Find the general solution of $\mathcal{L}(x) = Ae^{-2t}$. $\diamond$

12. [Barbălat's Lemma] Let $\varphi: \mathbb{R} \rightarrow \mathbb{R}$ be a C^1 function such that there exist the limits $\lim_{t \rightarrow \infty} \varphi(t)$ and $\lim_{t \rightarrow \infty} \varphi'(t)$ and both are finite. Prove that $\lim_{t \rightarrow \infty} \varphi'(t) = 0$. $\diamond$

13. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a C^1 function and $\varphi: \mathbb{R} \rightarrow \mathbb{R}$ be a solution of the differential equation $\dot{x} = f(x)$ such that $\lim_{t \rightarrow \infty} \varphi(t) = 2$. Prove that 2 is an equilibrium point of $\dot{x} = f(x)$. $\diamond$

14. Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ and $H: \mathbb{R}^2 \rightarrow \mathbb{R}$ be two C^1 functions such that H is a first integral in $\mathbb{R}^2$ of the planar system $\dot{X} = f(X)$. Prove that $\dot{X} = f(X)$ has no global attractor. $\diamond$